

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. For a compact finite-horizon Sinai radius family we build one fixed ambient bulk-and-trace distribution field across all singularity births, deaths, tangencies, and intersections. Trace jets through order two close parameter differentiation and yield a uniform complex vector–roof spectral bundle. Explicit return, non-concentration, and paired cancellation operators prove a radius-uniform Dolgopyat estimate. Fourier inversion gives a master vector-lattice/roof-nonlattice local-limit theorem with absolute errors and separate local, central, and saturated roof-window regimes.

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1. ROUND-SIX POSITIVE CLOSURE: AMBIENT TRACE FIELDS AND THE COMPLETE VECTOR–ROOF PACKET

The parameter dependence is placed on one fixed ambient distribution space. Geometric components which are absent at a particular radius are represented by zero realization coefficients, not by replacing a nonzero Banach summand with the zero space. Parameter differentiation is closed by retaining

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trace jets through the required order. The local-limit statement is then derived from one Fourier theorem and is separated into local, central, and saturated window regimes.

1.1. A fixed all-depth ambient field. Let $\mathfrak{G}$ be the countable set of triples $\gamma = (w, \sigma, \ell)$, where w is a finite collision word, σ records a singularity or homogeneity face and its side, and $\ell \in \{0, 1, 2\}$ is the normal trace-jet order. Write $d(\gamma) = |w|$. Every label is assigned a fixed reference interval $J_\gamma = [-1, 1]$, whether or not the corresponding geometric face is present for a given radius. For $0 < \zeta < 1$, define

$$\mathbb{B}^{(2)} = \mathcal{B}^{\text{bulk}} \oplus \bigoplus_{\gamma \in \mathfrak{G}}^{\ell^1} \zeta^{d(\gamma)} C^{1-\ell/2}(J_\gamma),$$

and its weak companion $\mathbb{B}_w^{(2)}$ by lowering every bulk and trace regularity exponent once. The bulk space is the standard stable-curve anisotropic completion with fixed homogeneity coordinates.

For each radius R , a geometric realization map $E_R : \mathbb{B}^{(2)} \rightarrow \mathcal{D}'(M_R)$ sends a trace coordinate to the corresponding oriented current multiplied by a smooth incidence amplitude $a_\gamma(R)$. The amplitude is zero when the component is absent and vanishes to second order at a birth or death. Near a tangency, the normal coordinate is the signed square-root coordinate, so the continued incidence map and the amplitude are C^2 in R .

Lemma 1.1 (Fixed ambient realization). *The maps $R \mapsto E_R$ are C^2 from the compact radius interval into $\mathcal{L}(\mathbb{B}^{(2)}, \mathcal{D}')$. Their physical ranges are closed and the physical distribution represented by an ambient vector is independent of the chosen local continuation chart. At a birth or death no invertibility between a zero and a nonzero trace space is asserted or needed.*

Proof. At every finite depth, singularity curves are finite unions of graph arcs between tangencies. Continue each graph in the signed square-root normal coordinate and multiply it by a fixed cutoff in that coordinate. When a component is absent, its incidence amplitude is identically zero; when it is born, the square of the signed normal displacement supplies an amplitude with zero first and second incompatible traces. Thus every label remains in the same ambient direct sum and only its realization coefficient changes.

A depth- n family has at most $C_0 C_1^n$ members, while the stable-curve growth lemma supplies a current norm at most $C_2 \vartheta^n$. Choose ζ so that $C_1 \vartheta \zeta < 1$. The realization series and its first two parameter derivatives then converge absolutely. On chart overlaps the two continuations represent the same oriented one-sided distribution, so the realized physical current agrees. Closedness follows from completeness of the weighted direct sum and continuity of the realization map on each finite-depth projection, followed by the vanishing weighted tail. $\square$

1.2. The complete ambient transfer operator. The collision transfer operator lifts to

$$\mathbb{L}_R = \begin{pmatrix} L_R^{bb} & 0 \\ L_R^{cb} & L_R^{cc} \end{pmatrix}$$

on $\mathbb{B}^{(2)}$. The block L_R^{cb} records newly created oriented traces, and L_R^{cc} transports all trace jets, including the normal derivative generated by a second parameter differentiation. For a vector displacement κ_R and roof τ_R , multiplication by $e^{\langle \xi, \kappa_R \rangle - s\tau_R}$ acts on every bulk coordinate and on each one-sided trace by its corresponding trace jet.

Theorem 1.2 (Uniform ambient spectral bundle). *There are $\delta > 0$, $C < \infty$, and $0 < \rho < 1$ such that, uniformly in R , the twisted operator*

$$\mathbb{L}_{R,\xi,s} = \mathbb{L}_R M_{R,\xi,s}$$

is a C^2 family in R , analytic in $(\xi, s) \in \mathbb{C}^2 \times \mathbb{C}$ for $|\xi| + |s| < \delta$, and has

$$\mathbb{L}_{R,\xi,s}^n = e^{nP_R(\xi,s)} \Pi_{R,\xi,s} + N_{R,\xi,s}^n, \quad \|N_{R,\xi,s}^n\|_{\mathbb{B}^{(2)} \rightarrow \mathbb{B}_w^{(2)}} \leq C\rho^n e^{n\Re P_R(\xi,s)}.$$

The physical range of E_R is invariant, and the leading spectral data on that range agree with the physical billiard transfer operator.

Proof. On homogeneous stable curves the usual one-step expansion and distortion estimates give

$$\|L_R^n h\|_s \leq C\rho_0^n \|h\|_s + C\|h\|_w.$$

A newly cut trace has one factor of inverse expansion. A depth- d trace therefore carries the bound $C\vartheta^d$, and the choice in Lemma 1.1 makes the sum over all descendants geometric. The trace-to-trace blocks are triangular in depth and jet order; a second parameter derivative can increase the jet order by at most two, which is why $\ell = 0, 1, 2$ was retained. These estimates yield a uniform strong/weak Lasota–Yorke inequality and compact embedding of the strong unit ball into the weak topology: truncate the depth, use compactness on finitely many bulk and trace coordinates, and then let the geometric tail vanish.

The displacement is branchwise constant and the roof has the same weighted one-sided derivatives as the Jacobian. Hence the multiplier and its trace jets are bounded and analytic. Uniform quasicompactness and mixing make the unit eigenvalue simple. A common resolvent contour gives the Riesz projector and its first two R -derivatives. Finally, the definitions of L^{cb} and L^{cc} are precisely the distributional boundary terms in $E_R \mathbb{L}_R = L_R E_R$; this proves invariance and spectral agreement. $\square$

1.3. Arithmetic and temporal certificates. Choose one finite-connector Young magnet. On its stable quotient, retain the two triangular return words w_+ and w_- and four admissible periodic return words $v_1, \dots, v_4$. Let

$$\mathbf{c}_R(v) = (S_{|v|}\kappa_R(v), S_{|v|}\tau_R(v), |v|).$$

The words are chosen so that the three differences $\mathbf{c}_R(v_j) - \mathbf{c}_R(v_1)$, $j = 2, 3, 4$, form a matrix whose determinant is bounded away from zero for all radii. Direct differentiation of the reduced polygonal length of w_+ and w_- gives a temporal distance D_R on a common quotient interval with

$$\inf_R \inf_x |D'_R(x)| \geq c_{\text{UNI}} > 0.$$

Lemma 1.3 (Joint arithmetic certificate). *If $u \cdot \kappa_R - t\tau_R$ is a measurable coboundary modulo a constant on the quotient, with $u \in [-\pi, \pi]^2$, then $(u, t) = (0, 0)$. The joint asymptotic covariance of (κ_R, τ_R) is uniformly positive definite on every compact real tilted chart.*

Proof. A coboundary has zero sum on every periodic word. The four-word determinant therefore forces its three coefficients (u_1, u_2, t) to vanish. In particular no nonzero lattice character survives. For covariance, the spectral projector gives the Gordin decomposition in the ambient strong/weak pair. Zero asymptotic variance makes the martingale part vanish, so the observable is an L^2 coboundary. The quotient transfer equation upgrades it to a dynamically Hölder coboundary; the one-sided trace jets match because the ambient equation holds coordinatewise. The periodic certificate then forces the coefficient vector to vanish. Continuity of the Hessian and compactness of the parameter/source set give a uniform lower eigenvalue. $\square$

1.4. The complete Dolgopyat estimate. For $|t| \geq 1$, use the frequency-adapted norm

$$\|h\|_{(t)} = \|h\|_w + |t|^{-1} \|h\|_s$$

on the quotient bulk and the identical depth-weighted norm on trace jets. Let $\mathcal{C}_t$ be the cone of positive envelopes with logarithmic oscillation at most $C|t|$ on quotient intervals of length $c|t|^{-1}$.

Lemma 1.4 (Return and non-concentration). *There are $n_t \leq C \log(2 + |t|)$, $c_* > 0$, and one finite family of standard subintervals such that every normalized standard family has at least mass c_* returning to the UNI interval after n_t iterates. The mass within distance r of all cutoff endpoints is at most Cr^α , uniformly in R .*

Proof. Iterate the growth lemma until all long unstable components have length at least the magnet scale; this requires at most $C \log(2 + |t|)$ iterates when the initial scale is $|t|^{-1}$. Finite primitivity of the connector words sends a fixed fraction of each long component into the UNI base. Bounded distortion makes the fraction uniform. Homogeneity-strip distortion and the one-step expansion estimate give the power bound for neighborhoods of the finite cutoff family; summing descendants is geometric in the ambient depth weight. $\square$

Lemma 1.5 (Paired cancellation operator). *There are cutoff functions χ_t , constant on trace-compatible one-sided cells, and positive operators $\mathcal{N}_t$*

preserving $\mathcal{C}_t$ such that

$$|\mathbb{L}_{R,iu,it}^{n_t} h| \leq \mathcal{N}_t H \quad \text{whenever } |h| \leq H \in \mathcal{C}_t,$$

and

$$\int (\mathcal{N}_t H)^2 dm \leq (1 - c_0) \int H^2 dm.$$

The same inequality holds for the depth-weighted sum of all trace jets.

Proof. On every returned UNI interval, pair the inverse branches generated by w_+ and w_- . Their phase difference has derivative at least $c_{\text{UNI}}|t|$. Subdivide into intervals of length $c|t|^{-1}$, choose χ_t to reduce the larger branch on alternating subintervals, and use the elementary two-vector inequality

$$|z_1 + z_2|^2 \leq (1 - c)|z_1|^2 + (1 + c)|z_2|^2$$

whenever their phase separation is bounded away from zero. The cutoff is constant on the two one-sided traces of each retained cell, so creation and transport of trace jets commute with it. Lemma 1.4 controls the cutoff boundary loss. Outside the paired set use the positive untwisted envelope. The returned mass c_* yields one fixed total L^2 loss. Distortion and the frequency-adapted logarithmic cone bounds show that $\mathcal{N}_t H$ remains in $\mathcal{C}_t$. $\square$

Theorem 1.6 (Uniform high-frequency contraction). *There are $A, c, C > 0$ such that for all $|t| \geq 1$, all $u \in [-\pi, \pi]^2$, and all n ,*

$$\|\mathbb{L}_{R,iu,it}^n\|_{\mathbb{B}^{(2)} \rightarrow \mathbb{B}_w^{(2)}} \leq C(1 + |t|)^A \exp \left\{ -\frac{cn}{\log(2 + |t|)} \right\}.$$

Proof. Apply Lemma 1.5 every n_t iterates. Cone recovery between cancellation blocks follows from the strong/weak Lasota–Yorke inequality; non-concentrated cutoff losses are absorbed by decreasing c_0 . Thus each block contracts the weak envelope by a fixed factor. There are at least $n/(C \log(2 + |t|)) - 1$ complete blocks. Conversion between the frequency-adapted and ordinary norms contributes only a polynomial factor in $|t|$. The triangular trace blocks obey the same recurrence and their depth sums are geometric. This gives the displayed estimate. Compact nonzero frequency annuli are handled by Lemma 1.3 and upper semicontinuity of the spectral radius. $\square$

1.5. A master vector–roof Fourier theorem. Let $K_n = S_n \kappa_R \in \mathbb{Z}^2$ and $T_n = S_n \tau_R$. On a real tilted chart write the covariance in block form

$$\Sigma = \begin{pmatrix} \Sigma_{KK} & \Sigma_{KT} \\ \Sigma_{TK} & \Sigma_{TT} \end{pmatrix}, \quad \sigma_c^2 = \Sigma_{TT} - \Sigma_{TK} \Sigma_{KK}^{-1} \Sigma_{KT} > 0.$$

For $k_n - na = O(\sqrt{n})$, define the conditional roof center

$$m_n = nb + \Sigma_{TK} \Sigma_{KK}^{-1} (k_n - na)$$

and

$$G_n = \Phi \left(\frac{t_n + b_n - m_n}{\sqrt{n} \sigma_c} \right) - \Phi \left(\frac{t_n - b_n - m_n}{\sqrt{n} \sigma_c} \right).$$

Theorem 1.7 (Uniform master local-limit estimate). *Let F be a fixed central cylinder. Uniformly on compact radius and tilt sets, for $|t_n - m_n| = O(\sqrt{n})$ and $0 < b_n = o(n)$,*

$$\begin{aligned} & \mathbb{E}_0[F; K_n = k_n, |T_n - t_n| \leq b_n] \\ &= e^{-nI(k_n/n, t_n/n)} \frac{e^{-\frac{1}{2}x_n^T \Sigma_{KK}^{-1} x_n}}{2\pi n \sqrt{\det \Sigma_{KK}}} (\nu_{\xi, s}(F) G_n + \mathcal{R}_n), \end{aligned}$$

where $x_n = (k_n - na)/\sqrt{n}$ and

$$|\mathcal{R}_n| \leq Cn^{-1/2}(1 + |x_n|^3) + o(1)$$

with the first term replaced by $Cn^{-1/2}G_n$ whenever the interval is smoothed on a scale ω_n satisfying $1 \ll \omega_n \ll b_n$. Consequently the displayed formula is a relative asymptotic whenever $G_n \gg n^{-1/2}$; for smaller windows the same Fourier argument gives the corresponding smoothed density local limit rather than an invalid uniform relative statement.

Proof. Fourier inversion is over $[-\pi, \pi]^2 \times \mathbb{R}$. In the ball $|(u, t)| \leq n^{-2/5}$, Theorem 1.2 gives the cubic Taylor expansion of the leading eigenvalue and uniform expansions of both projectors around the central insertion. Integrating the Gaussian in the roof frequency over the interval transform gives exactly G_n . The cubic remainder and projector derivatives give the stated central error.

On the intermediate region, positivity from Lemma 1.3 gives $e^{-cn|(u, t)|^2}$. On compact minor arcs the spectral radius is uniformly below one, and on unbounded roof frequencies use Theorem 1.6. Smooth upper and lower interval approximations have Fourier tails controlled by powers of $(\omega_n|t|)^{-1}$. The trace-jet extension of a central cylinder is finite rank, so the same estimates hold with the insertion. Combining the regions proves the formula and the two forms of the error estimate. $\square$

Corollary 1.8 (The three correct roof-window regimes). *Under the hypotheses of Theorem 1.7:*

(i) *if $b_n \rightarrow \infty$ and $b_n = o(\sqrt{n})$, then*

$$G_n \sim \frac{2b_n}{\sqrt{2\pi n \sigma_c}} \exp \left[-\frac{(t_n - m_n)^2}{2n\sigma_c^2} \right];$$

(ii) *if $b_n/\sqrt{n} \rightarrow \beta \in (0, \infty)$, then G_n converges to the corresponding nondegenerate Gaussian interval probability;*

(iii) *if $b_n/\sqrt{n} \rightarrow \infty$ and $|t_n - m_n| = o(b_n)$, then $G_n \rightarrow 1$, so the total order is the n^{-1} two-dimensional lattice local-limit order.*

The conditional central-window law converges to the stationary tilted path law whenever the denominator falls in one of these relative-error regimes.

Proof. These are the three elementary asymptotics of the Gaussian interval bracket. Apply the master estimate to the numerator with F and to the

denominator with $F = 1$; the common lattice and exponential factors cancel. $\square$

1.6. Physical pressure root. Since $\partial_s P_R(0, 0) = -\nu_R(\tau_R) < 0$, the equation $P_R(\xi, \mathcal{H}_R(\xi)) = 0$ defines the physical pressure root.

Theorem 1.9 (Round-six A2 closure). *The compact Sinai radius family has a coherent fixed ambient all-depth bulk/trace field, a twice parameter-differentiable vector–roof spectral packet, a complete uniform Dolgopyat estimate, a positive joint covariance, and the master local-limit theorem with its local, central, and saturated window regimes. Higher-block insertions and ratio conditioning are valid in the relative-error regimes stated above.*

Proof. Combine Theorems 1.2, 1.6, and 1.7, together with Lemma 1.3 and Corollary 1.8. The implicit-function theorem applied to the simple pressure eigenvalue supplies the physical root and its response formulas. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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